

Unit 4. Linear Maps and Matrix Representations

1. Linear maps, kernels, and images

All spaces in this unit are finite-dimensional real linear spaces. A **linear map** $T : V \rightarrow W$ satisfies

$$T(ax + by) = aT(x) + bT(y) \quad (x, y \in V, a, b \in \mathbb{R}).$$

Here V is the domain and W the codomain. Taking $a = b = 0$ gives $T(0) = 0$; repeated addition then gives

$$T\left(\sum_{j=1}^k c_j v_j\right) = \sum_{j=1}^k c_j T(v_j),$$

including the empty sum. The zero map $x \mapsto 0$ and identity map $I_V : x \mapsto x$ satisfy the defining equality directly. For $A : m \times n$, $T_A : \mathbb{R}^n \rightarrow \mathbb{R}^m$, $x \mapsto Ax$, is linear by the distributive laws. More generally, fixed $A : m \times p$, $B : q \times n$ give the linear map $X \mapsto AXB$ from $\mathbb{R}^{p \times q}$ to $\mathbb{R}^{m \times n}$, since

$$A(ax + bY)B = aAXB + bAYB.$$

Define

$$\ker T = \{x \in V : T(x) = 0\}, \quad \text{im } T = T(V) = \{T(x) : x \in V\}.$$

The map is **injective** if $T(x) = T(y)$ implies $x = y$, and **surjective** if $\text{im } T = W$. In the matrix case, $\ker T_A = N(A)$ and $\text{im } T_A = \mathcal{C}(A)$.

Proposition 4.1. For a linear map $T : V \rightarrow W$:

- (a) $\ker T$ and $\text{im } T$ are subspaces of V and W .
- (b) $T(\text{span}(v_1, \dots, v_k)) = \text{span}(T(v_1), \dots, T(v_k))$.
- (c) T is injective if and only if $\ker T = \{0\}$.

Proof. • Both sets contain zero. If $x, y \in \ker T$, then $T(ax + by) = 0$. If $u = T(x), v = T(y)$, then $au + bv = T(ax + by) \in \text{im } T$. These are the subspace tests.

- As $c_1, \dots, c_k$ range over $\mathbb{R}$, the identity

$$T\left(\sum_j c_j v_j\right) = \sum_j c_j T(v_j)$$

gives every vector on each side of (b). Empty lists give $\{0\}$.

- An injective map has only one preimage of zero, namely zero. Conversely, $T(x) = T(y)$ gives $x - y \in \ker T$; if that kernel is $\{0\}$, then $x = y$.

□

Theorem 4.2 (Prescribing the images of a basis). Let $B = (v_1, \dots, v_n)$ be a basis of V and let $w_1, \dots, w_n$ be any vectors of W . There is exactly one linear map $T : V \rightarrow W$ such that $T(v_j) = w_j$.

Proof. • Each $v \in V$ has unique coefficients $v = \sum_j x_j v_j$. Define $T(v) = \sum_j x_j w_j$. Uniqueness of the coefficients makes this a well-defined map.

- If $v = \sum_j x_j v_j$, $u = \sum_j y_j v_j$, the coefficients of $av + bu$ are $ax_j + by_j$. Hence

$$T(av + bu) = \sum_j (ax_j + by_j) w_j = aT(v) + bT(u).$$

In particular $T(v_j) = w_j$.

- Any linear map with these basis images must send $\sum_j x_j v_j$ to $\sum_j x_j w_j$, so it equals T .

For $V = \{0\}$, the empty basis prescribes the unique map $0 \mapsto 0$. $\square$

Thus two linear maps agreeing on a basis agree everywhere. More generally, agreement on a spanning list suffices: expand any vector in that list and apply linearity to each summand.

Theorem 4.3 (Rank–nullity). For every linear map $T : V \rightarrow W$,

$$\dim V = \dim \ker T + \dim \operatorname{im} T.$$

The number $\dim \operatorname{im} T$ is the **rank** of T , written $\operatorname{rank} T$.

Proof. • Extend a basis $(z_1, \dots, z_s)$ of $\ker T$ to a basis $(x_1, \dots, x_r, z_1, \dots, z_s)$ of V , using Unit 3, Theorem 3.3.

- Write any $v \in V$ as $v = \sum_i a_i x_i + \sum_j b_j z_j$. Then $T(v) = \sum_i a_i T(x_i)$; hence the $T(x_i)$ span $\operatorname{im} T$.
- Suppose $\sum_i c_i T(x_i) = 0$. Then $\sum_i c_i x_i \in \ker T$, so $\sum_i c_i x_i = \sum_j d_j z_j$. Independence of the extended basis forces every c_i, d_j to be zero. Thus $T(x_1), \dots, T(x_r)$ are independent.
- They form an image basis. Consequently $\dim \operatorname{im} T = r$, $\dim \ker T = s$, $\dim V = r + s$.

The argument includes either empty list and the zero domain. $\square$

A linear bijection is an **isomorphism**.

- Its inverse is linear: if $y = T(x), y' = T(x')$, then

$$T^{-1}(ay + by') = T^{-1}(T(ax + bx')) = ax + bx'.$$

- If (v_j) is a basis of V , surjectivity gives $w = T(v) = \sum_j c_j T(v_j)$ for every $w \in W$. Thus the images span W . For independence,

$$\sum_j c_j T(v_j) = 0 \implies T\left(\sum_j c_j v_j\right) = 0 \implies \sum_j c_j v_j = 0 \implies c_j = 0 \quad (\forall j).$$

Here the last two implications use injectivity and the original basis. An isomorphism therefore sends a basis to a basis, so $\dim V = \dim W$.

- Conversely, given bases $(v_j), (w_j)$ of the same length, Theorem 4.2 constructs $T(v_j) = w_j$. Every $\sum_j c_j w_j$ has the unique preimage $\sum_j c_j v_j$, so T is an isomorphism.

Proposition 4.4. If $\dim V = \dim W$, a linear map $T : V \rightarrow W$ is injective if and only if it is surjective.

Proof. By Proposition 4.1 and rank–nullity,

$$T \text{ injective} \iff \dim \ker T = 0 \iff \dim \operatorname{im} T = \dim V = \dim W.$$

Since $\operatorname{im} T \subseteq W$, the last equality is equivalent to $\operatorname{im} T = W$, by the equal-dimension subspace criterion of Unit 3. $\square$

Example 4.1 (Symmetrization). On $\mathbb{R}^{n \times n}$, $n \geq 1$, define

$$P(A) = \frac{A + A^\top}{2}.$$

- (a) Prove linearity. (b) Find $\ker P$. (c) Find its dimension. Also determine $\operatorname{im} P$ and $\operatorname{rank} P$.

(a) Transpose is linear, so

$$P(aA + bB) = \frac{aA + bB + aA^\top + bB^\top}{2} = aP(A) + bP(B).$$

(b) $P(A) = 0$ is exactly $A^\top = -A$: the kernel is the skew-symmetric space. (c) Such matrices have zero diagonal and $a_{ji} = -a_{ij}$. Thus

$$A = \sum_{i < j} a_{ij}(E_{ij} - E_{ji}).$$

These matrices are independent by their entries above the diagonal. They form a basis with $n(n-1)/2$ members: the $n(n-1)$ ordered distinct pairs occur in reversal pairs. Every $P(A)$ is symmetric; conversely, a symmetric S satisfies $P(S) = S$. Therefore the image is the symmetric space and

$$\text{rank } P = n^2 - \frac{n(n-1)}{2} = \frac{n(n+1)}{2}.$$

2. Coordinates and matrix representations

For an **ordered basis** $B = (v_1, \dots, v_n)$, the coordinate column $[v]_B$ is defined by

$$v = \sum_j x_j v_j, \quad [v]_B = (x_1, \dots, x_n)^\top.$$

The coordinate map $v \mapsto [v]_B$ is a linear bijection onto $\mathbb{R}^n$. Indeed, coefficients of $av + bu$ are $a[v]_B + b[u]_B$; equal coordinate columns give equal vectors, and every $x \in \mathbb{R}^n$ is the coordinate column of $\sum_j x_j v_j$. The order is part of the basis data. For the zero space, $\mathbb{R}^0$ contains just the empty coordinate column. The unique 0×0 matrix is I_0 ; its product with itself is I_0 , so $I_0^{-1} = I_0$. Products with empty inner index sets use the empty-sum convention of Unit 3.

Theorem 4.5 (Matrix of a linear map). Let $B = (v_1, \dots, v_n)$ and $C = (w_1, \dots, w_m)$ be ordered bases of V, W . For a linear map $T : V \rightarrow W$, put

$$A = [T]_{C \leftarrow B} = \begin{bmatrix} [T(v_1)]_C & \cdots & [T(v_n)]_C \end{bmatrix} \in \mathbb{R}^{m \times n}.$$

Then $[T(v)]_C = A[v]_B$ for every $v \in V$. This identity determines A uniquely. Conversely, every $m \times n$ matrix is the matrix of exactly one linear map in these bases.

Proof. • Write $T(v_j) = \sum_i a_{ij} w_i$ and $v = \sum_j x_j v_j$. Then

$$T(v) = \sum_j x_j \sum_i a_{ij} w_i = \sum_i \left(\sum_j a_{ij} x_j \right) w_i.$$

Uniqueness of the C -coefficients gives $[T(v)]_C = Ax$.

- Applying the identity to v_j , whose coordinate column is e_j , determines each column of A .
- Given A , prescribe $T(v_j) = \sum_i a_{ij} w_i$. Theorem 4.2 gives existence and uniqueness.

□

Coordinates also identify the two kernels and the two images:

$$\{[v]_B : v \in \ker T\} = N(A), \quad \{[w]_C : w \in \text{im } T\} = \mathcal{C}(A).$$

For the first equality, $T(v) = 0$ is equivalent to $A[v]_B = 0$, and every coordinate column has a preimage v . For the second, each image coordinate is $A[v]_B$; conversely, any Ax is the coordinate column of $T(\sum_j x_j v_j)$. The restricted coordinate maps are linear bijections. Since such maps preserve bases, they preserve dimension, giving

$$\text{rank } T = \text{rank } A, \quad \dim \ker T = \dim N(A).$$

Example 4.2. Suppose $B = (v_1, v_2)$, $C = (w_1, w_2, w_3)$ are ordered bases of V, W , and a linear map $T : V \rightarrow W$ satisfies

$$T(v_1) = 3w_1 - w_2 + 17w_3, \quad T(v_2) = w_1 + w_2 - w_3.$$

Find the representing matrix and the image of $x_1v_1 + x_2v_2$.

The coefficients of the two displayed images are the two columns:

$$[T]_{C \leftarrow B} = \begin{pmatrix} 3 & 1 \\ -1 & 1 \\ 17 & -1 \end{pmatrix}.$$

Consequently

$$T(x_1v_1 + x_2v_2) = (3x_1 + x_2)w_1 + (-x_1 + x_2)w_2 + (17x_1 - x_2)w_3.$$

For maps $T, S : V \rightarrow W$, define $(aT + bS)(v) = aT(v) + bS(v)$. If both maps are linear, expansion at $\alpha u + \beta v$ gives

$$(aT + bS)(\alpha u + \beta v) = \alpha(aT + bS)(u) + \beta(aT + bS)(v).$$

Thus their linear combination is linear; its values on basis members give $[aT + bS]_{C \leftarrow B} = a[T]_{C \leftarrow B} + b[S]_{C \leftarrow B}$.

Proposition 4.6 (Composition and inverses). Let $T : V \rightarrow W$ and $S : W \rightarrow U$ be linear, with representing matrices $A = [T]_{C \leftarrow B}$ and $D = [S]_{E \leftarrow C}$. The composition $ST : v \mapsto S(T(v))$ is linear, and

$$[ST]_{E \leftarrow B} = DA.$$

Also $[I_V]_{B \leftarrow B} = I_n$. If T is an isomorphism,

$$[T^{-1}]_{B \leftarrow C} = A^{-1}.$$

Proof. • Successive linearity gives $ST(av + bu) = S(aT(v) + bT(u)) = aST(v) + bST(u)$.

- For every v , $[ST(v)]_E = D[T(v)]_C = DA[v]_B$. Uniqueness of the representing matrix gives the formula.
- The identity sends v_j to v_j , giving the columns e_j . The inverse map is linear, as already proved. Its matrix H satisfies $HA = I_n$ and $AH = I_m$ from $T^{-1}T = I_V$, $TT^{-1} = I_W$. An isomorphism preserves dimension, so $m = n$, and $H = A^{-1}$.

□

Theorem 4.7 (Changing bases). Keep $A = [T]_{C \leftarrow B}$. Let $E = (u_1, \dots, u_n)$ and $F = (z_1, \dots, z_m)$ be new bases of V, W . Set

$$R = [I_V]_{B \leftarrow E}, \quad Q = [I_W]_{C \leftarrow F}.$$

Then R, Q are invertible, and

$$[v]_B = R[v]_E, \quad [w]_C = Q[w]_F, \quad [T]_{F \leftarrow E} = Q^{-1}AR.$$

Proof. • The first two identities are the representation formula applied to identity maps. Composing the coordinate changes in reverse order gives I_n or I_m ; hence both changes are invertible.

- For every $v \in V$,

$$Q[T(v)]_F = [T(v)]_C = A[v]_B = AR[v]_E.$$

Multiplying by Q^{-1} gives the claimed matrix.

□

For $T : V \rightarrow V$, using the same basis on domain and codomain gives the special rule

$$[T]_{E \leftarrow E} = R^{-1}[T]_{B \leftarrow B}R.$$

Matrices A, H are called **similar** if $H = R^{-1}AR$ for an invertible R . Every such R is a change-of-basis matrix: define u_j by $[u_j]_B = Re_j$. Its columns form a basis of $\mathbb{R}^n$, and the inverse coordinate map sends that basis to (u_j) . Thus similar matrices represent the same linear map in different bases. In particular they have equal rank, by the coordinate rank equality above.

Example 4.3 (A change of coordinates). In $\mathbb{R}^3$, take the ordered lists

$$B = ((1, 1, 0)^\top, (-1, 1, 1)^\top, (0, 1, 2)^\top), \quad C = ((2, 1, 1)^\top, (0, 0, 1)^\top, (-1, 1, 1)^\top).$$

Verify that they are bases and find $P = [I]_{C \leftarrow B}$.

Let U, V contain the columns of B, C :

$$U = \begin{pmatrix} 1 & -1 & 0 \\ 1 & 1 & 1 \\ 0 & 1 & 2 \end{pmatrix}, \quad V = \begin{pmatrix} 2 & 0 & -1 \\ 1 & 0 & 1 \\ 1 & 1 & 1 \end{pmatrix}.$$

If $U(a, b, c)^\top = 0$, its first and third equations give $a = b$, $b = -2c$; the second gives $a + b + c = -3c = 0$. Thus all three coefficients are zero, and the columns of U are a basis. For any $u = (u_1, u_2, u_3)^\top$, solve $Vp = u$:

$$2p_1 - p_3 = u_1, \quad p_1 + p_3 = u_2, \quad p_1 + p_2 + p_3 = u_3.$$

These have the unique solution

$$p_1 = (u_1 + u_2)/3, \quad p_2 = u_3 - u_2, \quad p_3 = (2u_2 - u_1)/3,$$

also proving that C is a basis. Substitute the three columns of U :

$$P = \begin{pmatrix} 2/3 & 0 & 1/3 \\ -1 & 0 & 1 \\ 1/3 & 1 & 2/3 \end{pmatrix}, \quad VP = U.$$

The direction is fixed by $[v]_C = P[v]_B$.

3. Restrictions, product ranks, and matrix equations

For a subspace $U \subseteq V$, the **restriction** $T|_U : U \rightarrow W$ is defined by $T|_U(x) = T(x)$ for $x \in U$. It is linear: $ax + by \in U$ and $T|_U(ax + by) = aT|_U(x) + bT|_U(y)$. Its kernel and image are

$$\ker(T|_U) = \{x \in U : T(x) = 0\} = U \cap \ker T, \quad \text{im}(T|_U) = \{T(x) : x \in U\} = T(U).$$

Theorem 4.8 (Rank of a composition). For linear maps $S : V \rightarrow W$, $T : W \rightarrow Z$,

$$\text{rank}(TS) = \text{rank } S - \dim(\text{im } S \cap \ker T).$$

In particular, for $A : m \times n$, $B : n \times p$,

$$\text{rank}(AB) = \text{rank } B - \dim(\mathcal{C}(B) \cap N(A)).$$

Proof. • Put $U = \text{im } S$. Then $\ker(T|_U) = U \cap \ker T$ and $\text{im}(T|_U) = T(S(V)) = \text{im}(TS)$.

• Rank-nullity for $T|_U$ gives

$$\text{rank}(TS) = \dim U - \dim(U \cap \ker T) = \text{rank } S - \dim(\text{im } S \cap \ker T).$$

- Take $S(x) = Bx$, $T(y) = Ay$ to obtain the matrix formula.

□

Proposition 4.9 (Product-rank bounds). For $A : m \times n$, $B : n \times p$,

$$\text{rank } A + \text{rank } B - n \leq \text{rank}(AB) \leq \min(\text{rank } A, \text{rank } B).$$

Moreover,

$$AB = 0 \iff \mathcal{C}(B) \subseteq N(A); \quad AB = 0 \implies \text{rank } A + \text{rank } B \leq n.$$

Proof. • The intersection in Theorem 4.8 has dimension between 0 and $\dim N(A) = n - \text{rank } A$. Substitution gives the lower bound and the upper bound $\text{rank}(AB) \leq \text{rank } B$.

- Every column combination $ABx = A(Bx)$ lies in $\mathcal{C}(A)$. Hence $\mathcal{C}(AB) \subseteq \mathcal{C}(A)$, giving the other upper bound.
- $AB = 0$ means that A annihilates every column of B . By linearity this is equivalent to annihilating their entire span. The final inequality follows by setting $\text{rank}(AB) = 0$ in the lower bound.

□

The exact formula also shows

$$\text{rank}(AB) = \text{rank } B \iff \mathcal{C}(B) \cap N(A) = \{0\}.$$

Indeed, equality holds exactly when the subtracted dimension is zero; a subspace has dimension zero exactly when it is $\{0\}$. The lower bound is called **Sylvester's inequality**.

Example 4.4. Must AB and BA have the same rank for square matrices A, B ? Consider

$$A = \begin{pmatrix} 1 & 0 \\ 1 & 0 \end{pmatrix}, \quad B = \begin{pmatrix} 0 & 0 \\ 1 & 1 \end{pmatrix}.$$

Direct multiplication gives

$$AB = 0, \quad BA = \begin{pmatrix} 0 & 0 \\ 2 & 0 \end{pmatrix}.$$

Their ranks are 0 and 1: the second matrix has one nonzero column. Thus their ranks can differ, although Unit 2 gives $\text{tr}(AB) = \text{tr}(BA)$.

Proposition 4.10 (Frobenius's inequality). For $A : m \times n$, $B : n \times p$, $C : p \times q$,

$$\text{rank}(AB) + \text{rank}(BC) \leq \text{rank } B + \text{rank}(ABC).$$

Proof. • Set $U = \mathcal{C}(B)$, $U' = \mathcal{C}(BC)$. Since $BCx = B(Cx)$, we have $U' \subseteq U$.

- Apply Theorem 4.8 twice:

$$\begin{aligned} \text{rank } B - \text{rank}(AB) &= \dim(U \cap N(A)), \\ \text{rank}(BC) - \text{rank}(ABC) &= \dim(U' \cap N(A)). \end{aligned}$$

- The inclusion $U' \cap N(A) \subseteq U \cap N(A)$ gives

$$\text{rank}(BC) - \text{rank}(ABC) \leq \text{rank } B - \text{rank}(AB).$$

Rearrange to obtain the stated inequality.

□

Theorem 4.11 (A matrix of right-hand sides). Let $A : m \times n$, $B : m \times q$, with $m, n, q \geq 1$, and let $X : n \times q$ be unknown. Then

$$AX = B \text{ is consistent} \iff \mathcal{C}(B) \subseteq \mathcal{C}(A) \iff \text{rank}[A \ B] = \text{rank } A.$$

If X_0 is a solution and the columns of $K : n \times (n - r)$ form a basis of $N(A)$, where $r = \text{rank } A$, all solutions are

$$X = X_0 + KZ, \quad Z \in \mathbb{R}^{(n-r) \times q}.$$

Different Z give different solutions. In particular, a consistent equation has a unique solution exactly when $\text{rank } A = n$.

Proof. • Writing $X = [x_1 \ \cdots \ x_q]$, $B = [b_1 \ \cdots \ b_q]$ gives $AX = B$ exactly when $Ax_j = b_j$ for every j . This is equivalent to all columns of B lying in $\mathcal{C}(A)$.

- Since $\mathcal{C}(A) \subseteq \mathcal{C}([A \ B])$, these spaces are equal exactly when their dimensions agree. Equality is equivalent to $\mathcal{C}(B) \subseteq \mathcal{C}(A)$.
- $AK = 0$, so $A(X_0 + KZ) = B$ for every Z . Conversely, $AX = B = AX_0$ gives $A(X - X_0) = 0$. Expand each column of $X - X_0$ in the basis columns of K ; collecting these coefficients gives $X - X_0 = KZ$.
- Independence of the columns of K gives $KZ = KZ' \Rightarrow Z = Z'$, by comparing columns. There are $(n - r)q$ independent scalar parameters. For $r = n$, the empty product KZ is zero. For $r < n$ and $q \geq 1$, changing a parameter gives a different solution.

□

Example 4.5. Given $A : 9 \times 7$ and $B : 4 \times 3$, prove that there is a nonzero 7×4 matrix X satisfying $AXB = 0$.

The map

$$\Phi : \mathbb{R}^{7 \times 4} \longrightarrow \mathbb{R}^{9 \times 3}, \quad X \longmapsto AXB$$

is linear by the distributive calculation at the start of the unit. The matrix-unit bases give dimensions 28 and 27. Since $\text{im } \Phi$ is a subspace of the codomain, $\text{rank } \Phi \leq 27$. Consequently

$$\dim \ker \Phi = 28 - \text{rank } \Phi \geq 1.$$

The kernel therefore contains a nonzero matrix, and every such matrix satisfies $AXB = 0$.

Exercises and solutions

Suggested practice: Exercises 4.2, 4.5, 4.8, 4.16, 4.21, and 4.25. All vector spaces below are finite-dimensional real spaces.

Linear maps and their kernels

Exercise 4.1 (Testing linearity). Determine which maps are linear. Give a proof or a counterexample in each case.

- (a) $F : \mathbb{R}^3 \rightarrow \mathbb{R}^2$, $F(x, y, z) = (x, z)^\top$.
- (b) $F : \mathbb{R}^4 \rightarrow \mathbb{R}^4$, $F(x) = -x$.
- (c) $F : \mathbb{R}^3 \rightarrow \mathbb{R}^3$, $F(x) = x + (0, -1, 0)^\top$.
- (d) $F : \mathbb{R}^2 \rightarrow \mathbb{R}^2$, $F(x, y) = (2x + y, y)^\top$.
- (e) $F : \mathbb{R}^2 \rightarrow \mathbb{R}^2$, $F(x, y) = (2x, y - x)^\top$.
- (f) $F : \mathbb{R}^2 \rightarrow \mathbb{R}^2$, $F(x, y) = (y, x)^\top$.
- (g) $F : \mathbb{R}^2 \rightarrow \mathbb{R}$, $F(x, y) = xy$.

Solution. The maps in (a), (b), (d), (e), and (f) are respectively multiplication by

$$\begin{pmatrix} 1 & 0 & 0 \\ 0 & 0 & 1 \end{pmatrix}, \quad -I_4, \quad \begin{pmatrix} 2 & 1 \\ 0 & 1 \end{pmatrix}, \quad \begin{pmatrix} 2 & 0 \\ -1 & 1 \end{pmatrix}, \quad \begin{pmatrix} 0 & 1 \\ 1 & 0 \end{pmatrix}.$$

Each is linear because $A(su + tv) = sAu + tAv$. In (c), $F(0) = (0, -1, 0)^\top \neq 0$, which rules out linearity. In (g), $F(1, 0) = F(0, 1) = 0$ but $F(1, 1) = 1$, so additivity fails.

Exercise 4.2 (Recovering a map from two values). In each case $L : \mathbb{R}^2 \rightarrow \mathbb{R}^2$ is linear. Find both $L(1, 0)$ and $L(0, 1)$.

- (a) $L(3, 1) = (1, 2)^\top$, $L(-1, 0) = (1, 1)^\top$.
- (b) $L(4, 1) = (1, 1)^\top$, $L(1, 1) = (3, -2)^\top$.
- (c) $L(1, 1) = (2, 1)^\top$, $L(-1, 1) = (6, 3)^\top$.

Solution. Express the standard columns using the given input columns, then apply L .

- (a) The identities

$$e_1 = -(-1, 0)^\top, \quad e_2 = (3, 1)^\top + 3(-1, 0)^\top$$

give $L(e_1) = (-1, -1)^\top$ and $L(e_2) = (4, 5)^\top$. They check the data: $3L(e_1) + L(e_2) = (1, 2)^\top$ and $-L(e_1) = (1, 1)^\top$.

- (b) Here

$$e_1 = \frac{1}{3}(4, 1)^\top - \frac{1}{3}(1, 1)^\top, \quad e_2 = -\frac{1}{3}(4, 1)^\top + \frac{4}{3}(1, 1)^\top.$$

Thus $L(e_1) = (-2/3, 1)^\top$, $L(e_2) = (11/3, -3)^\top$. The combinations $4L(e_1) + L(e_2)$ and $L(e_1) + L(e_2)$ are the two prescribed outputs.

- (c) Finally,

$$e_1 = \frac{1}{2}(1, 1)^\top - \frac{1}{2}(-1, 1)^\top, \quad e_2 = \frac{1}{2}(1, 1)^\top + \frac{1}{2}(-1, 1)^\top.$$

Hence $L(e_1) = (-2, -1)^\top$, $L(e_2) = (4, 2)^\top$. Their sum and their difference $L(e_2) - L(e_1)$ recover the given outputs.

Exercise 4.3 (Combining scalar-valued maps). Let $f, g : V \rightarrow \mathbb{R}$ be linear. Prove that $F : V \rightarrow \mathbb{R}^2$, $F(v) = (f(v), g(v))^\top$, is linear. Generalize to any finite number of scalar-valued linear maps.

Solution. For linear $f_1, \dots, f_p : V \rightarrow \mathbb{R}$, $p \geq 1$, define $F(v) = (f_1(v), \dots, f_p(v))^\top$. For $u, v \in V$ and $a, b \in \mathbb{R}$, coordinate j gives

$$f_j(au + bv) = af_j(u) + bf_j(v) \quad (1 \leq j \leq p).$$

Equality in every coordinate proves $F(au + bv) = aF(u) + bF(v)$. The requested pair is the case $p = 2$.

Exercise 4.4 (Independent images). Let $F : V \rightarrow W$ be linear. Suppose $w_1, \dots, w_k$ are independent and $F(v_i) = w_i$ for every i . Prove that $v_1, \dots, v_k$ are independent.

Solution. If $\sum_i a_i v_i = 0$, applying F gives $\sum_i a_i w_i = 0$. Independence of the w_i gives $a_i = 0$ for every i . This proves the required independence without assuming that F is injective on all of V .

Exercise 4.5 (A scalar-valued map). Let $F : V \rightarrow \mathbb{R}$ be linear, let $W = \ker F \neq V$, and fix $v_0 \notin W$.

- (a) Show that every $v \in V$ can be written $v = w + cv_0$, where $w \in W$ and $c \in \mathbb{R}$.
- (b) Prove that W is a subspace. If $(v_1, \dots, v_r)$ is a basis of W , prove that $(v_0, v_1, \dots, v_r)$ is a basis of V .

Solution.

- (a) Since $F(v_0) \neq 0$, set

$$c = \frac{F(v)}{F(v_0)}, \quad w = v - cv_0.$$

Then $F(w) = F(v) - cF(v_0) = 0$, so $w \in W$ and the formula holds.

- (b) We have $F(0) = 0$, and for $u, w \in W$, $F(au + bw) = aF(u) + bF(w) = 0$; hence W is a subspace. Part (a) and the basis of W prove spanning. In a relation $a_0 v_0 + \sum_{i=1}^r a_i v_i = 0$, applying F gives $a_0 F(v_0) = 0$, hence $a_0 = 0$. Independence in W gives all remaining coefficients zero. The argument also covers $W = \{0\}$, when its basis is empty.

Exercise 4.6 (Extending a linear map). Let $U \subseteq V$ be a subspace and let $S : U \rightarrow W$ be linear. Show that there is a linear $T : V \rightarrow W$ whose restriction to U is S .

Solution.

- Choose a basis $(u_1, \dots, u_r)$ of U and extend it to $(u_1, \dots, u_r, v_1, \dots, v_s)$ of V .
- Choose any $w_1, \dots, w_s \in W$, and define

$$T \left(\sum_{i=1}^r a_i u_i + \sum_{j=1}^s b_j v_j \right) = \sum_{i=1}^r a_i S(u_i) + \sum_{j=1}^s b_j w_j.$$

Unique basis coefficients make this well-defined; addition and scalar multiplication of the coefficients prove linearity.

- For $u \in U$, all $b_j = 0$, so $T(u) = S(u)$. If $U = V$ there are no new images to choose; if $U = \{0\}$, the first sum is empty. The new images w_j can be chosen freely.

Exercise 4.7 (A subspace avoiding the kernel). Let $T : V \rightarrow W$ be linear and $U \subseteq V$ a subspace satisfying $U \cap \ker T = \{0\}$. Let $u_1, \dots, u_r \in U$ be independent.

- (a) Prove that $T(u_1), \dots, T(u_r)$ are independent.
- (b) If also $r = \dim U$ and $U \oplus \ker T = V$, prove that these images form a basis of $\operatorname{im} T$.

Solution.

- (a) A relation $\sum_i a_i T(u_i) = 0$ gives

$$\sum_i a_i u_i \in U \cap \ker T = \{0\}.$$

Independence of the u_i forces every $a_i = 0$.

- (b) The u_i form a basis of U . For $v \in V$, write $v = u + z$ with $u \in U$, $z \in \ker T$, and then $u = \sum_i c_i u_i$. Thus $T(v) = T(u) = \sum_i c_i T(u_i)$, proving spanning. Part (a) proves independence, including the empty list if $T = 0$.

Exercise 4.8 (Factoring through another map). Let $T : V \rightarrow W$ and $R : U \rightarrow W$ be linear. If $\operatorname{im} T \subseteq \operatorname{im} R$, prove that there is a linear $S : V \rightarrow U$ such that $T = RS$.

Solution.

- Choose a basis $(v_1, \dots, v_n)$ of V . For each i , choose $u_i \in U$ with $R(u_i) = T(v_i)$, which is possible by the inclusion of images.
- Define $S(\sum_i a_i v_i) = \sum_i a_i u_i$. Unique coefficients make S well-defined; their addition and scaling give linearity. For every $v = \sum_i a_i v_i$,

$$RS(v) = \sum_i a_i R(u_i) = \sum_i a_i T(v_i) = T(v).$$

Thus $T = RS$. If $V = \{0\}$, the zero map S gives the same conclusion.

Exercise 4.9 (Two invertible maps). Show that each of the following maps $L : \mathbb{R}^3 \rightarrow \mathbb{R}^3$ is invertible.

- (a) $L(x, y, z) = (x - 2y, x + z, x + y + 2z)^\top$.
- (b) $L(x, y, z) = (2x - y + z, x + y, 3x + y + z)^\top$.

Solution. Both maps are linear by their coordinate formulas. Solve $L(x, y, z) = (u, v, w)^\top$ for arbitrary u, v, w .

- (a) From $x = u + 2y$ and $z = v - x$, the last equation becomes

$$w = x + y + 2(v - x) = 2v - u - y.$$

Hence

$$y = 2v - u - w, \quad x = 4v - u - 2w, \quad z = u - 3v + 2w.$$

These equations determine the only possible preimage. Substitution into the three coordinates gives u, v, w , so it exists for every output.

- (b) The second equation gives $y = v - x$, and the first gives $z = u + v - 3x$. The last equation is then $w = u + 2v - x$. Thus the unique candidate is

$$x = u + 2v - w, \quad y = -u - v + w, \quad z = -2u - 5v + 3w.$$

Substitution gives $2x - y + z = u$, $x + y = v$, and $3x + y + z = w$. Again every output has exactly one preimage.

Exercise 4.10 (A nonzero map whose square is zero). Let $L : \mathbb{R}^2 \rightarrow \mathbb{R}^2$ be linear, $L \neq 0$, and $L^2 = L \circ L = 0$. Show that there is a basis (a, b) of $\mathbb{R}^2$ such that $L(a) = b$ and $L(b) = 0$.

Solution. Since $L \neq 0$, choose a with $b = L(a) \neq 0$. Then $L(b) = L^2(a) = 0$. If $sa + tb = 0$, applying L gives $sb = 0$, hence $s = 0$. The original relation becomes $tb = 0$, so $t = 0$. These two independent columns form the required basis of $\mathbb{R}^2$.

Exercise 4.11 (An idempotent map). Let $P : V \rightarrow V$ be linear and satisfy $P^2 = P$. Prove

$$V = \ker P + \operatorname{im} P, \quad \ker P \cap \operatorname{im} P = \{0\}.$$

Solution. For $v \in V$,

$$v = (v - Pv) + Pv, \quad P(v - Pv) = Pv - P^2v = 0.$$

The first term lies in $\ker P$, the second in $\operatorname{im} P$; therefore their sum is all of V . If w belongs to the intersection, write $w = Pv$. Then

$$w = Pv = P^2v = P(w) = 0.$$

This proves that the sum is direct.

Exercise 4.12 (A left inverse and a right inverse). Let $T : V \rightarrow W$, $R : W \rightarrow V$, and $S : W \rightarrow V$ be maps satisfying $RT = I_V$ and $TS = I_W$.

- (a) Show that T is invertible.
- (b) Show that $R = S = T^{-1}$.

Solution.

- (a) If $T(v) = T(v')$, applying R gives $v = v'$. For $w \in W$, the element $S(w)$ satisfies $T(S(w)) = w$. Thus T is both injective and surjective.
- (b) For every $w \in W$,

$$R(w) = R(T(S(w))) = S(w).$$

The identities $RT = I_V$, $TS = I_W$ therefore make this common map the inverse of T .

The proof requires no linearity assumption on these maps.

Coordinates and matrix representations

Exercise 4.13 (Reading a representation from basis images). Find the indicated representation matrices.

- (a) $F : \mathbb{R}^4 \rightarrow \mathbb{R}^2$, $F(x_1, x_2, x_3, x_4) = (x_1, x_2)^\top$, using standard bases.
- (b) $F : \mathbb{R}^4 \rightarrow \mathbb{R}^4$, $F(x_1, x_2, x_3, x_4) = (x_1, x_2, 0, 0)^\top$, using standard bases.
- (c) A linear map $L : V \rightarrow V$, with a basis $B = (v_1, \dots, v_n)$ and $L(v_i) = c_i v_i$, using B for both domain and codomain.

Solution. Column j is the coordinate column of the image of input basis vector j . Thus the matrices are respectively

$$\begin{pmatrix} 1 & 0 & 0 & 0 \\ 0 & 1 & 0 & 0 \\ 0 & 0 & 0 & 0 \\ 0 & 0 & 0 & 0 \end{pmatrix}, \quad \begin{pmatrix} 1 & 0 & 0 & 0 \\ 0 & 1 & 0 & 0 \\ 0 & 0 & 0 & 0 \\ 0 & 0 & 0 & 0 \end{pmatrix}, \quad \operatorname{diag}(c_1, \dots, c_n).$$

For (c), $[L(v_j)]_B = c_j e_j$, which checks every diagonal and off-diagonal entry. The first two maps have different codomains and hence different numbers of matrix rows.

Exercise 4.14 (Changing coordinates). In $\mathbb{R}^3$, let

$$B = \left((3, 2, 1)^\top, (0, -2, 5)^\top, (1, 1, 2)^\top \right), \quad C = \left((1, 1, 0)^\top, (-1, 2, 4)^\top, (2, -1, 1)^\top \right).$$

Find $[I]_{C \leftarrow B}$, the matrix converting B -coordinates to C -coordinates.

Solution. Let U, V have the columns of B, C , respectively:

$$U = \begin{pmatrix} 3 & 0 & 1 \\ 2 & -2 & 1 \\ 1 & 5 & 2 \end{pmatrix}, \quad V = \begin{pmatrix} 1 & -1 & 2 \\ 1 & 2 & -1 \\ 0 & 4 & 1 \end{pmatrix}.$$

If $U(a, b, c)^\top = 0$, the first two rows give $c = -3a$ and $b = -a/2$; the third then gives $-15a/2 = 0$. Thus all three coefficients vanish, so B is a basis. To express $(r, s, t)^\top$ in C , solve

$$a - b + 2c = r, \quad a + 2b - c = s, \quad 4b + c = t.$$

Subtracting the first equation from the second and using the third gives

$$b = \frac{s - r + 3t}{15}, \quad c = \frac{4r - 4s + 3t}{15}, \quad a = \frac{2r + 3s - t}{5}.$$

The unique solution for every (r, s, t) also proves that C is a basis. Apply this formula to the three columns of U :

$$[I]_{C \leftarrow B} = P = \begin{pmatrix} 11/5 & -11/5 & 3/5 \\ 2/15 & 13/15 & 2/5 \\ 7/15 & 23/15 & 2/5 \end{pmatrix}.$$

The product $VP = U$ verifies all three coordinate expansions. For example its first column is $(11/5)(1, 1, 0)^\top + (2/15)(-1, 2, 4)^\top + (7/15)(2, -1, 1)^\top = (3, 2, 1)^\top$.

Exercise 4.15 (Constructing a desired coordinate change). Let $B = (v_1, \dots, v_n)$ be a basis of V , and let $A \in \mathbb{R}^{n \times n}$ be invertible. Write $A^{-1} = (f_{ij})$ and set

$$w_j = \sum_{i=1}^n f_{ij} v_i \quad (1 \leq j \leq n).$$

Prove that $C = (w_1, \dots, w_n)$ is a basis of V and $[I]_{C \leftarrow B} = A$.

Solution. A relation $\sum_j c_j w_j = 0$ has B -coordinate column $A^{-1}c$. Thus $A^{-1}c = 0$, so $c = 0$. The n independent vectors form a basis of the n -dimensional space. For any $v \in V$, expanding its C -coordinates in the v_i gives

$$[v]_B = A^{-1}[v]_C, \quad [v]_C = A[v]_B.$$

This is exactly the claimed representation of the identity.

Exercise 4.16 (Changing both bases). Let $T : \mathbb{R}^4 \rightarrow \mathbb{R}^3$ be

$$T(x) = (x_1 + x_2, x_2 + x_3 - x_4, x_1 - x_3 + x_4)^\top.$$

Let B, C be the standard bases of $\mathbb{R}^4, \mathbb{R}^3$, and let the ordered bases E, F be

$$\begin{aligned} E &= \left((1, -1, 0, -1)^\top, (0, 0, 1, 1)^\top, (0, 0, 0, 1)^\top, (1, 1, 0, 0)^\top \right), \\ F &= \left((1, 0, 1)^\top, (1, 1, 0)^\top, (-1, 0, 0)^\top \right). \end{aligned}$$

- Find $[T]_{C \leftarrow B}$.
- Find both $[I]_{B \leftarrow E}$ and $[I]_{F \leftarrow C}$.
- Find $[T]_{F \leftarrow E}$ by each of two methods: directly express the images of the E basis in F ; then use the change-of-basis formula with the matrices in (a) and (b).
- Find $\text{rank } T$ and $\dim \ker T$.

Solution. Write $F = (f_1, f_2, f_3)$.

- The images of the four standard columns give

$$A = [T]_{C \leftarrow B} = \begin{pmatrix} 1 & 1 & 0 & 0 \\ 0 & 1 & 1 & -1 \\ 1 & 0 & -1 & 1 \end{pmatrix}.$$

- The first matrix has the columns of E :

$$R = [I]_{B \leftarrow E} = \begin{pmatrix} 1 & 0 & 0 & 1 \\ -1 & 0 & 0 & 1 \\ 0 & 1 & 0 & 0 \\ -1 & 1 & 1 & 0 \end{pmatrix}.$$

Its columns are independent: $Rc = 0$ gives $c_1 + c_4 = -c_1 + c_4 = 0$, so $c_1 = c_4 = 0$; rows 3 and 4 then give $c_2 = c_3 = 0$. For $y = (y_1, y_2, y_3)^\top$, writing $y = af_1 + bf_2 + cf_3$ gives

$$(a + b - c, b, a)^\top = y, \quad (a, b, c) = (y_3, y_2, -y_1 + y_2 + y_3).$$

Consequently

$$Q = \begin{pmatrix} 1 & 1 & -1 \\ 0 & 1 & 0 \\ 1 & 0 & 0 \end{pmatrix}, \quad [I]_{F \leftarrow C} = Q^{-1} = \begin{pmatrix} 0 & 0 & 1 \\ 0 & 1 & 0 \\ -1 & 1 & 1 \end{pmatrix}.$$

These coordinate equations check $Q^{-1}Q = QQ^{-1} = I_3$.

- Denote the four members of E by $u_1, \dots, u_4$. Directly, their images are

$$T(u_1) = 0, \quad T(u_2) = 0, \quad T(u_3) = (0, -1, 1)^\top = f_1 - f_2, \quad T(u_4) = (2, 1, 1)^\top = f_1 + f_2,$$

Thus

$$H = [T]_{F \leftarrow E} = \begin{pmatrix} 0 & 0 & 1 & 1 \\ 0 & 0 & -1 & 1 \\ 0 & 0 & 0 & 0 \end{pmatrix}.$$

For the second method, multiplication gives

$$AR = \begin{pmatrix} 0 & 0 & 0 & 2 \\ 0 & 0 & -1 & 1 \\ 0 & 0 & 1 & 1 \end{pmatrix}, \quad Q^{-1}AR = \begin{pmatrix} 0 & 0 & 1 & 1 \\ 0 & 0 & -1 & 1 \\ 0 & 0 & 0 & 0 \end{pmatrix} = H.$$

This also verifies $QH = AR$, the four direct coordinate expansions.

- (d) In H , columns 3 and 4 are independent: $a + b = 0$, $-a + b = 0$ imply $a = b = 0$. All other columns are zero. By the coordinate rank identity following Theorem 4.5, $\text{rank } T = \text{rank } H = 2$, and Theorem 4.3 gives $\dim \ker T = 4 - 2 = 2$.

Exercise 4.17 (Bases adapted to a map). Let $T : V \rightarrow W$ be linear of rank $k > 0$, with $\dim V = n$ and $\dim W = m$. Find bases E, F for which

$$[T]_{F \leftarrow E} = \begin{pmatrix} I_k & 0 \\ 0 & 0 \end{pmatrix}.$$

Solution.

- Choose a basis $z_1, \dots, z_{n-k}$ of $\ker T$ and extend it to a basis $E = (x_1, \dots, x_k, z_1, \dots, z_{n-k})$ of V .
- The proof of Theorem 4.3 shows that $T(x_1), \dots, T(x_k)$ form a basis of $\text{im } T$: every image is a combination of them, and a zero combination lifts to a relation between the x_i and the kernel basis, forcing all coefficients zero.
- Extend this image basis to a basis F of W . The first k input basis vectors have output coordinate columns $e_1, \dots, e_k$; all remaining input basis vectors map to zero. These columns give the displayed matrix, of size $m \times n$.

Exercise 4.18 (Subspaces preserved by a map). Let $T : V \rightarrow V$ be linear, $\dim V = n$. A subspace U is *invariant under T* when $T(U) \subseteq U$.

- (a) If U is invariant and $\dim U = r$, show that there is a basis B with

$$[T]_{B \leftarrow B} = \begin{pmatrix} G & H \\ 0 & K \end{pmatrix}, \quad G \in \mathbb{R}^{r \times r}.$$

- (b) If $U \oplus W = V$ and both subspaces are invariant, find a basis B with

$$[T]_{B \leftarrow B} = \text{diag}(G, K), \quad G \in \mathbb{R}^{r \times r}, \quad r = \dim U.$$

Solution.

- (a) Extend a basis $u_1, \dots, u_r$ of U to a basis B of V . For $j \leq r$, the image $T(u_j)$ is in U , so its last $n - r$ coordinates in B vanish. These are precisely the lower left block entries. The remaining entries form the unrestricted blocks G, H, K .
- (b) Take a basis of U followed by a basis of W . The combined list spans V . A zero combination gives an element in $U \cap W = \{0\}$, so both separate combinations are zero; their basis independence proves that the combined list is a basis. Images of the U vectors have zero W -coordinates, and images of the W vectors have zero U -coordinates. Both off-diagonal blocks vanish. If a subspace is zero, its basis and corresponding blocks are empty, and the same argument applies.

Ranks of products and matrix equations

Exercise 4.19 (Multiplication by a full-rank matrix). For all compatible real matrices, prove

- (a) $\text{rank}(BA) = \text{rank } A$ if B has full column rank.
- (b) $\text{rank}(AC) = \text{rank } A$ if C has full row rank.

Solution.

- (a) Write $B : p \times m$, $A : m \times n$. The equation $By = 0$ has only $y = 0$, since the columns of B are independent. If $u_1, \dots, u_r$ is a basis of $\mathcal{C}(A)$, then $Bu_1, \dots, Bu_r$ span $\mathcal{C}(BA)$. Moreover, $\sum_i a_i Bu_i = 0$ implies $B(\sum_i a_i u_i) = 0$, then $\sum_i a_i u_i = 0$, and finally all $a_i = 0$. Thus the two column spaces have equal dimensions.
- (b) Write $A : m \times n$, $C : n \times q$. Full row rank gives $\mathcal{C}(C) = \mathbb{R}^n$. Every column Ax in $\mathcal{C}(A)$ therefore has $x = Cy$ for some y , so $Ax = ACy \in \mathcal{C}(AC)$. The reverse inclusion follows from $ACy = A(Cy)$. The column spaces are equal, so their dimensions are equal.

Exercise 4.20 (Removing a middle factor). For $n \times n$ matrices A, B, C , is it always true that $\text{rank}(ABC) \leq \text{rank}(AC)$?

Solution. No. Set

$$A = \begin{pmatrix} 1 & 0 \\ 0 & 0 \end{pmatrix}, \quad B = \begin{pmatrix} 0 & 1 \\ 1 & 0 \end{pmatrix}, \quad C = \begin{pmatrix} 0 & 0 \\ 0 & 1 \end{pmatrix}.$$

Then $AC = 0$, whereas

$$BC = \begin{pmatrix} 0 & 1 \\ 0 & 0 \end{pmatrix}, \quad ABC = \begin{pmatrix} 0 & 1 \\ 0 & 0 \end{pmatrix}.$$

Thus $\text{rank}(AC) = 0$ and $\text{rank}(ABC) = 1$, disproving the assertion.

Exercise 4.21 (A zero corner block). Let $A : m \times n$, $B : m \times p$, $C : q \times n$, and $D : q \times p$. Prove both inequalities

$$\text{rank} \begin{pmatrix} A & B \\ C & 0 \end{pmatrix} \geq \text{rank } B + \text{rank } C, \quad \text{rank} \begin{pmatrix} 0 & B \\ C & D \end{pmatrix} \geq \text{rank } B + \text{rank } C.$$

Solution. Choose independent columns $b_{j_1}, \dots, b_{j_s}$ forming a basis of $\mathcal{C}(B)$, and columns $c_{i_1}, \dots, c_{i_r}$ forming a basis of $\mathcal{C}(C)$. In the first block matrix, select the columns

$$\begin{pmatrix} b_{j_\ell} \\ 0 \end{pmatrix} \quad (1 \leq \ell \leq s), \quad \begin{pmatrix} a_{i_h} \\ c_{i_h} \end{pmatrix} \quad (1 \leq h \leq r).$$

In a zero combination, the lower block forces all coefficients of the second list to vanish. The upper block then forces the coefficients of the first list to vanish. Thus there are $r + s$ independent columns.

For the second block matrix, use

$$\begin{pmatrix} 0 \\ c_{i_h} \end{pmatrix} \quad (1 \leq h \leq r), \quad \begin{pmatrix} b_{j_\ell} \\ d_{j_\ell} \end{pmatrix} \quad (1 \leq \ell \leq s).$$

Now the upper block first forces the second list's coefficients to vanish; the lower block then deals with the first list. Again there are $r + s$ independent columns. The argument includes empty lists when either rank is zero.

Exercise 4.22 (A block proof of the product inequalities). (a) Using Exercise 4.21, prove for $A : m \times n$, $B : n \times p$, $C : p \times q$ that

$$\text{rank}(AB) + \text{rank}(BC) \leq \text{rank } B + \text{rank}(ABC).$$

(b) Deduce that for $A : m \times p$, $B : p \times n$, $\text{rank}(AB) \geq \text{rank } A + \text{rank } B - p$.

(c) With the sizes in (b), deduce that $AB = 0$ implies $\text{rank } A \leq p - \text{rank } B$.

Solution.

(a) Put

$$Z = \begin{pmatrix} 0 & AB \\ BC & B \end{pmatrix}, \quad P = \begin{pmatrix} I_m & -A \\ 0 & I_n \end{pmatrix}, \quad Q = \begin{pmatrix} I_q & 0 \\ -C & I_p \end{pmatrix}.$$

The matrices obtained from P, Q by replacing $-A, -C$ with A, C are their inverses: in either multiplication order the off-diagonal blocks cancel and the diagonal blocks remain identities. Block multiplication gives

$$PZ = \begin{pmatrix} -ABC & 0 \\ BC & B \end{pmatrix}, \quad PZQ = \begin{pmatrix} -ABC & 0 \\ 0 & B \end{pmatrix}.$$

For any block diagonal matrix $\text{diag}(H, K)$, a basis of its column space is obtained by embedding a basis of $\mathcal{C}(H)$ in the upper coordinates and a basis of $\mathcal{C}(K)$ in the lower coordinates. These columns span; a zero combination, read in the two blocks separately, has every coefficient zero. Therefore its rank is $\text{rank } H + \text{rank } K$. Invertible multiplication preserves rank, and multiplication by -1 preserves a column space. Hence

$$\text{rank } Z = \text{rank}(ABC) + \text{rank } B.$$

Exercise 4.21 also gives $\text{rank } Z \geq \text{rank}(AB) + \text{rank}(BC)$, proving (a).

(b) Apply (a) to the three factors A, I_p, B :

$$\text{rank } A + \text{rank } B \leq p + \text{rank}(AB).$$

(c) In (b), substitute $\text{rank}(AB) = 0$ and rearrange.

Exercise 4.23 (Using the rank of the middle factor). Let $A, X, B \in \mathbb{R}^{n \times n}$. Prove

$$\text{rank}(AXB) \geq \text{rank}(AX) + \text{rank}(XB) - \text{rank } X$$

by using a rank factorization of X .

Solution.

- If $X = 0$, every term is zero. Otherwise let $r = \text{rank } X > 0$ and choose $X = UV$, with $U : n \times r$ of full column rank and $V : r \times n$ of full row rank.
- Exercise 4.19 gives

$$\text{rank}(AX) = \text{rank}(AUV) = \text{rank}(AU), \quad \text{rank}(XB) = \text{rank}(UVB) = \text{rank}(VB).$$

- The factors $AU : n \times r$ and $VB : r \times n$ have inner size r . Applying Proposition 4.9 to them gives

$$\text{rank}(AXB) = \text{rank}((AU)(VB)) \geq \text{rank}(AU) + \text{rank}(VB) - r,$$

which is the required inequality.

Exercise 4.24 (When cancellation is valid). (a) Let $A : m \times n$, $C : n \times q$, and $B : q \times p$. Prove both statements:

$$\begin{aligned} \text{rank}(AC) = \text{rank } C &\implies \mathcal{R}(ACB) = \mathcal{R}(CB), & \text{rank}(ACB) = \text{rank}(CB), \\ \text{rank}(CB) = \text{rank } C &\implies \mathcal{C}(ACB) = \mathcal{C}(AC), & \text{rank}(ACB) = \text{rank}(AC). \end{aligned}$$

(b) Let $A, B : m \times n$. Prove that

$$\begin{aligned} C : r \times q, D : q \times m, \text{rank}(CD) = \text{rank } D, CDA = CDB &\implies DA = DB, \\ C : n \times q, D : q \times p, \text{rank}(CD) = \text{rank } C, ACD = BCD &\implies AC = BC. \end{aligned}$$

Solution.

(a) Rows of a product are combinations of the rows of its right factor. Thus $\mathcal{R}(AC) \subseteq \mathcal{R}(C)$. Under the first rank hypothesis these spaces have equal dimensions, hence are equal. Each row of C can therefore be expressed using the rows of AC ; collecting those coefficients gives a matrix $L : n \times m$ with $C = LAC$. Now

$$CB = LACB, \quad ACB = A(CB),$$

so each of $\mathcal{R}(CB)$ and $\mathcal{R}(ACB)$ is contained in the other. Their equality gives equality of ranks. Under the second hypothesis, $\mathcal{C}(CB) \subseteq \mathcal{C}(C)$ and equal dimensions give equality. Express the columns of C using the columns of CB : $C = CBR$ for some $R : p \times q$. The identities

$$AC = ACBR, \quad ACB = (AC)B$$

give both inclusions of column spaces and hence both claimed equalities.

(b) Put $F = A - B$. For the first assertion, the first rank statement in (a), applied to the factors C, D, F , gives

$$\text{rank}(DF) = \text{rank}(CDF) = 0,$$

since $CDF = CDA - CDB = 0$. Therefore $DF = 0$, or $DA = DB$. For the second assertion, use the second rank statement in (a) with factors F, C, D :

$$\text{rank}(FC) = \text{rank}(FCD) = 0.$$

Thus $FC = 0$, giving $AC = BC$.

Exercise 4.25 (Parameters and consistency). For real α, β , consider

$$\begin{aligned} \alpha x_1 + \beta x_2 + 2x_3 &= 1, \\ \alpha x_1 + (2\beta - 1)x_2 + 3x_3 &= 1, \\ \alpha x_1 + \beta x_2 + (\beta + 3)x_3 &= 2\beta - 1. \end{aligned}$$

When is the system consistent? When is its solution unique?

Solution. Subtract the first equation from the second and from the third. Then replace the first by its difference with the new second equation. These reversible operations give

$$\begin{aligned} \alpha x_1 + x_2 + x_3 &= 1, \\ (\beta - 1)x_2 + x_3 &= 0, \\ (\beta + 1)x_3 &= 2(\beta - 1). \end{aligned}$$

No parameter has been divided out.

- If $\beta = -1$, the last equation is $0 = -4$, so there is no solution.
- If $\beta = 1$, the last two equations give $x_3 = 0$, and the first gives $x_2 = 1 - \alpha x_1$. Thus

$$(x_1, x_2, x_3) = (t, 1 - \alpha t, 0) \quad (t \in \mathbb{R}),$$

so there are infinitely many solutions for every α .

- Suppose $\beta \neq -1, 1$. Back substitution gives

$$x_3 = \frac{2(\beta - 1)}{\beta + 1}, \quad x_2 = -\frac{2}{\beta + 1}, \quad \alpha x_1 = \frac{5 - \beta}{\beta + 1}.$$

If $\alpha \neq 0$, this determines the unique solution

$$(x_1, x_2, x_3) = \left(\frac{5 - \beta}{\alpha(\beta + 1)}, -\frac{2}{\beta + 1}, \frac{2(\beta - 1)}{\beta + 1} \right).$$

If $\alpha = 0$, consistency requires $\beta = 5$. For $\alpha = 0, \beta = 5$, all solutions are $(t, -1/3, 4/3)$, $t \in \mathbb{R}$; for the other values in this case there is no solution.

The displayed families satisfy the transformed equations, so reversing the row operations verifies them in the original system. Thus uniqueness holds exactly when $\alpha \neq 0$ and $\beta \neq -1, 1$.